Unpacking How Pole Dancers Come to Experience the Use of Fitness Trackers

Whitney-Jocelyn Kouaho
University of California Irvine
Irvine, California, USA
wkouaho@uci.edu

Daniel A. Epstein
Informatics
University of California, Irvine
Irvine, California, USA
epstein@ics.uci.edu

Abstract

Personal informatics literature has typically examined how fitness trackers can support understanding of cardio-based exercises, like running, which are closely associated with the step counts central to these devices. Although this focus has supported individuals in gleaning fruitful data discoveries, it confines the manner in which a body moves to these movement domains. In this paper, we contend with pole dance, a physical activity whose movement affordances (e.g., rotational, artistic, feminine) differ greatly from those centralized in the self-tracking technologies. We qualitatively interviewed 20 polers, and gathered their reflection on pole dance and how they view the use self-tracking devices for the activity. Using their insights on the inherent dynamicity of poling, independently, and upon interaction with fitness trackers, we offer suggestions to our discipline to challenge ourselves to re-imagine sensing as a pluralistic endeavor.

CCS Concepts

• **Human-centered computing**; • **Social and professional topics** → *Women*;

Keywords

personal informatics, pole dance, fitness trackers, feminism

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1 Introduction

In personal informatics research, physical activity is the most widely-studied domain [21]. Much of the excitement surrounding fitness tracking technologies at the heart of this work comes from the purported possibility for their use as supplemental health tools that sense the body, display suggestions, and ultimately provide (data) clarity on a user's "fitness" status. This helps in part to explain why interest in these devices has seen a steady incline over the last two decades and the use of them is expected to continue to grow exponentially [62].

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Yet, although there is an emphasis on the capacity for trackers to engender complex activity insights, in research using *commercial fitness tracking devices* specifically, there is a dearth of studies which focus on non-ambulatory physical activities [37]. More commonly, researchers have investigated the use of commercial fitness trackers as they pertain to activities which can be captured by steps, which is conceivable considering that most devices contain accelerometer sensors. And although designers have broadened the types of activities accounted for on commercial fitness trackers, represented through "modes," such as on a device like the Fitbit Charge 6 which has over forty that users can select [1], devices tend to return normative data between modes. As it pertains to the capacity for fitness trackers to engage with activities outside of these typical conventions, there are limitations to our understanding. As much of our thinking in this realm centers activities which engage movements optimal for collecting step count data, there has not been much exploration into whether personal informatics design goals are being adequately addressed in use contexts outside of the norm. Pointedly, a large contingency of studies have focused on the use of trackers by the general population, with studies who looked at specific populations primarily investigating use by patients, students, and office workers, with a small percentage of studies evaluating use in hobby spaces [21]. To that end, we attempt to offer an inquiry into the use of fitness trackers for the practice of recreational pole dance.

Recreational pole dance is a rich area to explore because there are many facets which when investigated in conjunction with the universalist model of current prevailing fitness tracking devices, can be useful for unpacking how fitness trackers interpret complex movement modalities. Of the activity "modes" available on commercial fitness tracking devices, pole dance is not one of them. Whether the reason that pole dance is excluded from this domain because it is, as a feminized activity, not considered "fitness" (in the personal informatics context) a sentiment akin to those continued contestations to categorize activities like ballet as sport because an activity cannot be both art & sport; note decades long failed attempts to introduce pole to the Olympics [2], or because of general designer uneasiness with clear embraces of sexuality and eroticism [10], which Osipova et al. posit is because of "sex exceptionalism", which places sex(uality) as an exceptional activity at the extreme ends of positive and negative associations which marginalize [49], is not easily reconciled.

Regardless, many scholars in HCI and related disciplines have already criticized how devices standardize and quantify body movement [15, 41, 57]. But we seek to interrogate pole in relationship to these circumstances particularly because it presses upon these

assumptions in three ways. Firstly, "steps" are limited, and movements in pole are fluid and varying. How does use operate in a scenario where design relies on its capacity to track movement that appears to be its foil? Secondly, in pole dance, the body and its fleshiness roots the practice. For example, it is heavily suggested that polers forego jewelry or certain clothing due to safety reasons [7]. However, automated tracking via sensing necessitates that wrist watches most commonly, or rings, be worn. How is this negotiated and understood? And how do devices respond to the inherent tension between design and body? Finally, as pole dance is culturally reconciled as a hyper-feminized activity, and considering the tension it poses to the masculinist ideals of universality embedded into the design of self-tracking technologies, we can engage in new conversations rooted in feminist HCI [11] and pluralist frameworks like soma design [31].

To carry out this inquiry, we interviewed 20 polers based in the United States, exploring both their general perspectives on pole as a movement framework and also on fitness tracking for pole. We situated our work with these research questions:

- RQ1:How do polers personally define movement for pole?
- RQ2: What are participant perspectives on the (physical) use of tracking devices while poling?
- RQ3: What are poler reactions to data that can be obtained from wearables in relation to pole dancing?

From our interviews, we found that dancers constantly negotiate between movement types which require different uses of the body such as strength or flexibility, and that body awareness impacts the way polers both move through space and understand their device use. We also found that although the fit of the watch can impact poling, some participants were able to find workarounds, while still articulating that the watch form is not optimal. Lastly, participants had varying reactions to the limitations of usable data. Some forego-ed use, others use the watch merely to keep a record of their activity, and others hack the data, creating value in the margins.

In the following sections, we wrestle with how these devices, as both subject and object, interact with polers, and how polers, as both subject and object, interact with devices. We reflect on these findings by bringing forward considerations for fitness tracking both for pole and generally, to re-evaluate how we think about *what* we sense and *where* we do that sensing, *why* we do that particular sensing, the physical form that sensing object will take, and *how* that data will be represented.

2 Background

Our investigation of polers perspectives on wearables and pole dance is influenced by the histories of pole dance and how its been understood in research, previous work in dance studios in HCI, and how embodied interaction has been conceptualized by soma design.

2.1 Pole dance

The physical demands of pole is understood to be quite intensive. Although the specificity of that intensity varies by sub genre of pole dance and their relative movement focus, generally, there are a set of interrelating physical dynamics which polers are beholden to. To secure their bodies on the pole, dancers must extensively train themselves to withstand their own weight, similar to resistance training [47], and even work to improve strength in targeted areas of the body such as their hands. By example, women who participated in pole had increased hand grip strength (increased as a poler was more advanced) in comparison to women who did not pole, at levels which are "higher than tennis players" [45].

Polers build up their strength in a myriad of ways, but one way this happens is through an active resistance of gravitational forces. Using their "hands and other surfaces of the body as brakes, [they] rely[ing] on the friction between skin and the pole" [54], to regulate movements, with the intensity of this "braking" further affected by factors such as whether the pole is static or spinning. Additionally, forms of pole which may be more "routine-based" or dance heavy, have been reported to require higher levels of cardio-respiratory exertion as opposed to pole aerial or "skills-based" classes or activities, as the former requires continuous movement and the latter is stop and start. Yet, generally, pole meets the qualifications to be designated at minimum a "moderate intensity cardio-respiratory exercise" in line with prevailing medical guidelines for daily exercise [47]. It has also been observed that polers have improved posture at levels similar to gymnastics and other forms of dance, increased muscle mass in upper body, and improved flexibility [45], with particular skills of "agility", "balance", and body control [47].

Although participating in pole does increase heart rate and caloric expenditure in line with data returns of commercial fitness trackers, in thinking on the physical practice of pole, we interrogate how these trackers understand not only the dynamic movements in pole which we will describe in section 4.1, but how they capture the complexity of the simultaneous nature of these sensations.

On the other hand, just as pole is physically demanding, it is also inherently *erotic*, as defined by Audre Lorde. By her estimation, the erotic is not necessarily a euphemism for *sexuality*. The erotic is a representation of that which is found in presence. To Lorde, "...the erotic is not a question only of what we do; it is a question of how acutely and fully we can feel in the doing," it is something that "we have been taught to suspect...warned against it all our lives by the male world, which values this depth of feeling enough to keep women around in order to exercise it in the service of men, but which fears this same depth too much to examine the possibilities of it within themselves" [40]. We emphasize that participating in pole dance activates the erotic sensibility to "feel." And although we focus on recreational pole dance in this paper, we do not make any attempt to distance pole from its erotic and sexual past or present.

Prior literature on pole dance has primarily been situated in gender studies, where there are evolving debates on whether pole dance is empowering or characteristic of patriarchal power structures [25, 65], discussions on social stigma of recreational pole [35], theories on the fetishization of polers [18], auto-ethnographies of social media shadow banning [7], of course feminist inquiries [28] and more.

These prevailing conversations stand to reason as the most recognized contemporary cultural understanding of the origins of pole dance is the twentieth century adult club industry. Even with the attempts beginning in the 1990's to disembody pole dance from its history as sex work, re-attach it to recreation, and by proxy, change its "ethics", such spaces and their apparent associations with sex work and femininity are often criticized as being hyper-sexualized,

demeaning, or even religiously offensive, by both feminist groups and those with disparate ideologies[28], page 28.

Of the very limited HCI selection of research on pole, one in particular by Toups Dugas and Tanenbaum opens new dialogues to build upon surrounding pole dance and embodiment [59]. In their work, an auto-ethnography of the first author's experiences taking pole dance classes over a span of seven months, during which she journaled and reflected upon her experiences with pole, brings forth a concept of "failing beautifully", or how poling invites continual processes of (physical, social, emotional, etc.) alignment and misalignment. In thinking about more expansive approaches for considering the trans experience in embodied interaction for interaction design, they suggest that a framework which highlights "failing beautifully" could enlighten experiences of the body under/not recognized by cis researchers and designers. We add to this contribution by thinking on how similar observations of the embodied characteristics of pole could be useful specifically for thinking on the design of fitness trackers.

2.2 Wearables, physical activity, and self tracking

Here, we introduce advances in leveraging wearables for physical activity tracking within the HCI community. We particularly highlight how these discussions have primarily focused on implementing novel sensing and suggesting opportunities for future implementations, or studying the motivations and practices of technology use. Less work has critically examined the activities and bodies that fall outside such frameworks, which we build on in the case of pole dancing.

HCI has long aimed to advance capacity for tracking and sensing of physical activity via wearable devices. Early research focused on leveraging sensors called accelerometers to detect steps and movement [9, 13], with further work moving to classify and track specific repetitive exercises [34, 44]. Recent contributions to this space have attempted to address areas where we can push sensing to further accommodate the complexities of the sports and activities we attempt to track. For example Zeagler, through an overview of the literature on wearables and their physical forms, contends that when designing trackers, researchers and designers should consider dynamics of the body in conjunction with the data they hope to collect [68]. To achieve this, among other dimensions, Zeagler points to a need to consider where the wearable is placed on the body("body placement"), where and how the body is sensitive to touch ("haptics"), how to recognize biometrics ("thermal tolerance"), and interpersonal comfort ("social acceptability").

Turmo Vidal et al. relatedly analyze the wearables design space in the sports context in order to provide a working framework for the development of tracking tools. In reviewing prior research in the domain, this work posits that device use in sports should aim to either enhance "existing parts of the practice" or enable "something new in the practice that could not be achieved without the wearable intervention". Yet, the capacity to enhance or enable "relates to the design goal of the interventions". They also identify that, 'input" or what is sensed, "output", how data is returned and the purpose it serves for the sport, are key dimensions which must be attended to in order to achieve these aims. The other two main areas of the design space are the combined relationship of input/output, and lastly wearability [61]. We use these frameworks to interpret participant perspectives and reflect on opportunity for wearables in the case of pole dancing.

Secondly, researchers have been invested in understanding how and why people use fitness wearables to self-track. Among other reasons, most often people self-track to achieve behavior change goals or for general record keeping [23]. Yet researchers have found that barriers exist to long term use, typically because of mismatches in expectations regarding what users look for in wearables technology [16, 22], inaccurate data [67], not useful data [38] and desires for improved aesthetics and fit of the device [29]. Others have pointed out that abandonment of wearables is frequent (usually within two months due to similar reasons as stated above) [38, 56]. While research has surfaced that wearables often do not support the types of activities people undertake [17, 29, 37], even when devices are specifically designed for the sports they were partaking in (e.g., sports equipment with sensors), people often do not find outputted data valuable because it does not align with the information they actually wanted to know [38]. By examining use and non-use of wearables in pole dancing, we surface more diverse tracking motivations and misalignments between goals and data.

Lastly, some research has aimed to use (novel) sensors to measure and track dance activities. Examples include tracking leg movements in jazz dance [58], motion sensing for social dance [19], assisting ballet and performance learning [60], or even wearable interfaces for changing music during sessions, [53]. These contributions are typically aiming to promote learning the dance activity, and rely on returning normative data related to performance for dance. We add onto this work both by focusing specifically on pole dance, but also in broadening our inquiry outside of the normative data output. Additionally we answer to the call by Volkova et al. stemming from their further studied use of and attitudes towards wearables in dance studios, within which they recommend that future work evaluate how policies on jewelry impact the potential to use wearables in the studio [64]. We examine such policies and uses in the case of pole dancing.

2.3 Pluralism, the body, and soma design

We offer some background into pluralism as a design philosophy, and how that approach manifests in soma design. In our study, we contend with pole dancing as a pluralist activity, and hypothesize that to adequately support its inclusion in self-tracking, we must fundamentally re-imagine how we design fitness wearables.

In personal informatics, research has often pointed out limits to wearables, particularly that the technologies can reinforce gendered design values [15] and societal norms around fitness [57], and exclude individuals whose behaviors and practices do not align with those norms [14, 37]. One underlying basis for these critiques stems from the disciplines' tendency towards designing for *universality*. These tendencies, which in some part show up in how we categorize movement into step count, calories, and heart rate, endeavor to capture a universal truth that is fundamental to all (imagined) "potential *users*" (a term Dourish & Mainwaring discourage)[20]. For some cardio-based activities like running or hiking, this framing of the body has a more direct one to one through line and on

the surface does not require much logical maneuvering, for other activities, the use of this measurement as the operandi is not as apparent.

Some could argue that these habits tend to thinking on "the body as anatomy and physiology" a description of attempts in HCI to "understand" bodily functions [39] through the use of existing (non-HCI) nomenclatures that categorize physiological and anatomical signals of bodies then digitally translates them into our designs. But even so, these design frameworks come up short in the diversity of *movement* that any one individual body or *bodies* as defined by Homewood et al. [30] can perform. This both strips the individual of the intricacies of their physical histories or everyday feel of their bodies, but also defines bodies as neutral and knowable in their neutrality. Thus, an individual step becomes evocative of a body politic[s], made clear by the very notion of the possibility of its standardization as a homogeneous representation of a bodies' *movement.*

Pluralism is considered one of the more useful ways to combat the impulses of universal design tendencies. In HCI, practitioners have considered the potential of various methodologies (with pluralist roots) for breaking through the hegemony, like generative method based feminism [6, 11].

One of those methodologies, which in some part builds onto those feminist HCI tools, is a framework coined, "soma design." This method, influenced by Ssoma-aesthetic theories, emphasizes how developing the awareness of ones body and its sensations [27], can be a useful tool for pluralistic design. Soma-aesthetics theories propose that to enrich our capacities to sense ourselves, we must engage in practices like Laban, meditation, and others, which open up our presence in mind-body connection. Soma design takes the central ideas in soma aesthetics and applies it directly to the design of technology [33], and in doing so " ...demands a non-dualistic stance, where the plurality of bodies becomes inescapable." Ultimately, soma design offers HCI a framework for rounding the limiting dichotomies at the center of our thinking.

For these aims, there are a set of soma design methods which have been suggested for achieving the goals of pluralism. Höök et al posit that one way we can invite pluralism into our designs is through first-person embodied explorations, where designers can feel their way to design insights [33]. They suggest that to engender insights reminiscent of the mind-body interaction, designers must use methods like these to evoke a deeper capacity to reflect on their senses and thus richer understanding of user experiences [31]. While we primarily report on the perspectives of a cohort of pole dancers, the first author leverages her first-person embodied experiences learning to pole dance as part of analysis. Höök et al also suggest that the inquiry should not end there however, they emphasize that a deeper somatic design method should be participatory, and should (somatically) engage users in the exploration[31].

Such methods are not only generative for bringing forth new ways of thinking about the body, these methods have been valuable for rethinking how data should be reflected and visualized. For example, Perovich & Zinzi engaged dancers in exercises as described above, to "find" data in their bodies. From this work, they developed a unique a concept which they call "data somatics", where they shifted from a focus on visual data representations towards data representations based in the body [50]. Hua et al. rethink form, or the physical, material, and shape, in design, through a first person, soma-aesthetic exploration of women's masturbation represented through embroidery [32]. Soma-aesthetic methods have even been used to design different haptics to represent heart beats [8] and wearable devices to be used in meditation [24]. We highlight these examples to emphasize that soma design offers us an opportunity to reconstitute many normative tendencies as they pertain to self-tracking, even in the way we represent data or consider form.

Although our study does not offer data gathered from embodied methods, we contend that pole dance, like yoga or meditation, is a somatic practice from which we can gather generative understandings about the practice of feeling. We offer explanations for the potential of pole for eliciting deep reflections, gathered from our inquiry with polers on their perspectives on pole dance as a physical activity and how the use of fitness trackers impacts that experience, and ultimately call attention to tracking concerns as they relate to form, data representation, and the felt experience.

3 Methods

To begin to develop an understanding of how pole dancers think about self tracking, we engaged a cohort 20 of pole dancers and instructors in conversations about their (non)use of fitness trackers.

3.1 Search & Inclusion Criteria

To identify potential participants, the first author began by searching a pole studio database which held information on Black-owned pole studios in the US. We felt it was important to begin recruitment from this population so as to ensure a study demographic which was not ethnically homogeneous, and also to center the population which is at the root of much of pole dance culture. From our search, we emailed over a dozen studios around the continental US to inquire about potential participation in our study. We also contacted pole studios in the research team's metropolitan area. In the end, we sent direct emails to 25 pole dance studios and 4 independent pole instructors, and all participants were either directly recruited by the research team or from snowball sampling.

In the recruitment email, we provided general information about the study procedures, contextualized the first authors' experience with pole dance and dance in general, expected participant compensation, and a link to a pre-study questionnaire with demographic questions. We also clarified that we wanted perspectives from participants who had both used and not used wearable devices.

To participate in the study, all participants needed to be at least 18, fluent in English, and either, taught pole dance, or pole danced recreationally, for at least six months. All interviews were conducted via Zoom and lasted for no more than an hour. Pole dance instructors were compensated $100, and all others were compensated $50 via a virtual payment service of their choosing (e.g., Zelle, Venmo, etc). We particularly focused on recruiting instructors because they had great subject matter expertise and could speak about the discipline both from their perspectives as practitioners and as observers of their own students.

In all we conducted 20 interviews.

Table 1 enumerates the profile of the participants we interviewed. Although not included in the table, we gathered data on participant salaries, length of time teaching, household makeup, location,

P (Gender, Age)	Ethnic background	Occupation(s)	Pole dance experience	Fitness watch(es) used
P1 (F, 42)	Latina	Social Worker, Pet Sitter	6 years	Fitbit
P2 (F, 39)	African American	Therapist	13 years	Apple Watch
P3 (F, 38)	Black	Social Work Manager	10 years	Garmin, Fitbit
P4 (F, 30)	Chinese	N/A	4 years	-
P5 (F, 29)	Taiwanese	Yoga Instructor, Tech Worker	2 years	Galaxy
P6 (F, 22)	Russian	Student	6 years	Apple Watch
P7 (F, 26)*	Chinese	Software Engineer	3 years	Garmin
P8 (F, 27)*	Chinese / Thai	Mechanical Engineer	3 years	-
P9 (F, 33)	Filipino	Program Manager	5 years	Apple Watch
P10 (F, 33)	White	Pharmaceudical Scientist	8 years	Appe Watch, Whoop
P11 (F, 36)	Mixed	International Banking	7 years	Fitbit
P12 (F, 25)*	Latina	N/A	2 years	-
P13 (M, 33)	Info-Fijian	Project Manager	5 years	Apple Watch
P14 (F, 26)	Irish + Italian	Editor	10 years	-
P15 (F, 28)	White/Caucasian	Barista, Freelance Photography	2 years	-
P16 (F, 39)	Mexican American	Business Owner, Strategy Development	11 years	Unspecified device
P17 (F, 39)	African American/Black	Education Coordinator, Pole Instructor	6 years	Whoop, Apple Watch
P18 (F, 33)	White/ Caucasian	Pole Studio Co-Owner & Manager	11 years	Fitbit, Oura Ring
P19 (F, 33)	Hispanic	Pole & Gymnastics Instructor, Marketing, Social Media	13 years	Whoop
P20 (F, 23)	Filipina/White	Fitness Studio Manager, Aerial Instructor, Kids Circus Coach	4 years	-

Table 1: Self-reported demographics of our 20 interview participants, of whom 17 were instructors and 3 were students. Participants had all been polers for at least 2 years, and 14 had used a fitness watch. Student polers are indicated with a *.

certification, and time spent poling in a week. To that effect, it is important to emphasize that our population skews middle class and all have either obtained a higher education degree, or were in the process of receiving one. Our populations educational and professional experiences inevitably come to bear on their perspectives on the use of fitness trackers, as has been documented by other researches within the personal informatics space [37, 48]. These perspectives tend to align more closely with class as its currently imbued in self trackers, which generally views fitness as leisure, champions independence instead of interdependence, and engage at a higher health literacy. Studies show that insights from our population would lean more towards device use.

Finally, participants had been practicing pole dancing for an average of 6.5 years (min 2, max 13). Instructors had been teaching for an average 4.8 years (min 6 months, max 13 years). P12 had never taught pole dance, and at the time of our interviews, P7 and P8 had each only taught for two months. All three were interviewed as student polers. 14 participants regularly wore a fitness wearable either within or outside the context of poling, while 6 did not. Aligning with pole dancing's rise in recreational popularity, only one of the interviewees had ever poled as an adult dancer. We cannot understate that the experiences of pole dancers in the sex work arena will differ.

3.2 Study Procedures

To allow for a flexible conversation, the interview was semi-structured. Although there was a general structure to the interview as it pertains to the larger "topic" sections, questions and their specificity were malleable to the interviewee and the conversations we were having.

That being considered, we thematically split the interview into two primary sections, the first, "General Pole Dancing" included questions about the practice of pole irrespective of device use. We thought to begin with this line of inquiry to both situate the participants experiences with poling, and also to gather their perspectives on unique manifestations of the practice such as the clothing limitations. The goal of this section was to get firsthand accounts on the intricacies of this practice as both fitness/physical and artistic medium. During the second part of the interviews, the "Fitness Tracker" section, we dove more deeply into participant's experiences with fitness tracking. The interviewees were made aware of these topic delineations, both at the beginning of the interview and upon "changing" sections.

3.2.1 General questions about pole dancing practice. We first engaged the polers in a line of inquiry about their history with pole

dance before delving more deeply into questions which directly supported our research questions. We began the study in this manner to both elucidate their histories with pole dance and fitness generally, and to contextualize these experiences with the responses they would later give regarding fitness trackers. This was also useful for settling the interviewee and giving them space to feel comfortable. This part of the interview included questions which contextualized the participants experience, such as *"What made you want to get into poling? Pole teaching?"*, to more general practice based inquiries like *"What can you tell me about the clothing restrictions in pole?"*, or inquiries about physicality to the effect of *"What to you, makes the 'fitness' part of pole dancing?"* We also asked questions to start to get at measures or metrics of performance, which is often what trackers optimize for such as *"How do you know if you've had a good session? What does good session mean to you?"*

3.2.2 Fitness tracker use. In this section, we dove deeply into participants experiences with, and their perspectives on, fitness trackers in their poling practice. Although many of the participants had used fitness trackers at some point (Table 1), some had discontinued use and others had never used fitness trackers. For those who had never used trackers, we maneuvered these conversations to explore more on their non-use and also engaged them in hypothetical inquiries about how trackers might assist or inhibit their pole practice. For those who had discontinued use, we both learned about their experiences when they used devices, the contexts of this use, and also the reasoning behind their discontinuation. We begin by asking participants to *"Tell me more about your experience with fitness trackers."* This gave us a whole picture of their device use outside of pole, before we transitioned to think about tracking for pole. Of course, for many participants their answers were fluid, and if they began to talk about pole tracking, we allowed them to steer the conversation in that direction. This was particularly generative for those polers who used trackers for other activities (e.g., hiking, the recreational gym,etc) but not for poling, because it helped to clarify why poling specifically dissuaded them from fitness tracker use.

This led well into discussions headed by questions like *"Have you ever considered using a fitness tracker for pole dancing?"*, or, *"What data insights do you look to gather when tracking for pole?"* In this section, all of our questions served to answer R2 and R3, thus we discussed participants physical experience with using trackers for pole, how they think about data, and the pursuit of progress in pole.

3.3 Analysis

All interviews were initially transcribed using Otter.Ai. The first author used these initial transcripts as a template, but manually corrected the materials (using the audio and video recordings as guides) to ensure that participant's experiences were accurately represented.

The first author begin the analysis by open coding the transcripts from the first two interviews. From this, we were able to develop 20 preliminary codes, which were used for subsequent interview analysis. As the interviews progressed and the questions shifted slightly, new codes were added, and by the end of the first round of analysis there were 36 codes. During a second round of coding, we re-analyzed the data and consolidated, removed, or maintained the earlier codes. As the authors had developed a more clear understanding of the perspectives gained from the interviews, we were able to prioritize certain details to be coded instead of others. In line with thematic analysis [12], we combined the codes to produce 3 themes and 7 sub-themes, which form the organization of our paper.

4 Findings

In this section, we first define and offer examples of some fundamental physical techniques in pole dancing. We then offer insights gleaned from our participants, unpacking how polers frame feeling as a key dimension of their practice (RQ1), and then trace perspectives on the use and non-use of self-tracking devices as it relates to both the physical experience of device use (RQ2) and the data experiences (RQ3). Quotes have been lightly edited for clarity.

4.1 Defining physical pole terms

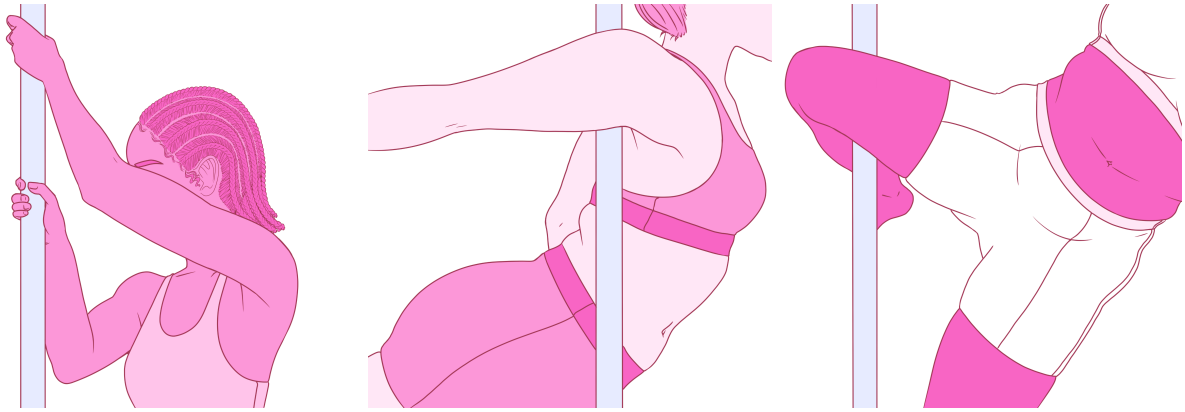
Pole is characterized by its complexity, often represented by its many subcategories (e.g. aerial pole, artistic pole, low flow, chair, combinations of all of these, and more). Within these different styles of pole, the exertion and categorization of movements vary some, such as more static tricks in an aerial class or fluid sensual movements in a burlesque class. Generally however, students are taught core techniques in "Introduction to Pole" classes, which most studios suggest students enroll in before broadening or advancing into other subcategories of pole.

These foundational techniques utilize physical fundamentals including *gripping, squeezing, and climbing*. A *"grip"* (or *"gripping"*) is how one attaches their body onto the pole [4]. One can grip using many parts of the body such as the hands, wrists, thighs, inner arms, glutes, pits of the elbows or knees, and more (Figure 1). Once in a grip, polers must contract their muscles, or *squeeze* them, to secure their bodies on the pole. Although squeezing requires muscle engagement, the success of these actions depends on details as granular as the placement, dryness or moistness, and amount of, skin in contact with the pole. Thus, n order to successfully support these techniques, polers must work to build and maintain *flexibility and strength*.

Without strength, one cannot sustain themselves long enough to maintain their grip, squeeze, or hold, and certainly cannot lift their bodies to perform fundamental movements like *climbs* and *inverts*. When a poler traverses the pole vertically, they are climbing. To perform a basic climb, polers wrap their legs around the pole and pull themselves upwards, activating a large set of muscles in their bodies to do so [3]. Similar to grips, there a myriad of climbs, and they are performed on poles of varying heights. *Inverts* are performed with the body upside-down on the pole (Figure 2) [5]. Often times, inverts are used as transitionary movements for getting into more advanced aerial tricks. Strength to sustain one's own body weight then, in combination with flexibility, heavily impacts one's ease with the activity.

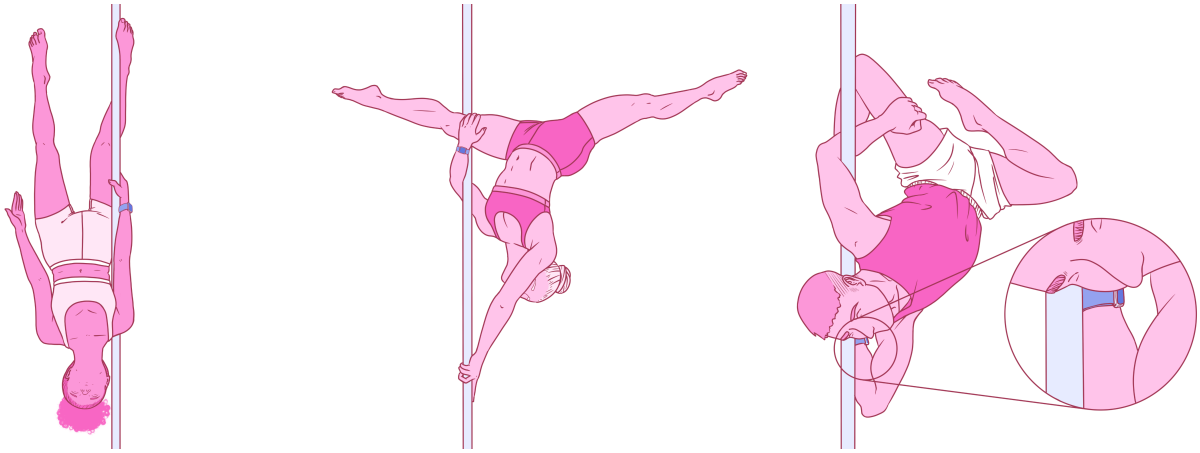
4.2 RQ1: Defining feeling in pole

We first highlight poler reflections that their practice is one that inherently relies on deep somatic engagement. We describe how this somatic interpretation occurs, and highlight how this awareness



(a) Anchor Grip **(b) Armpit Grip** **(c) Kneepit Grip**

Figure 1: Some forms of *grips* used in pole dancing. To successfully grip, one must *squeeze* their muscles to secure their bodies to the pole, such as with their hands, or around one of their joints. Sketches drawn by Carolina Arias Arguedas.



(a) Batman **(b) Cheba Split** **(c) Broken Pegasus**

Figure 2: Three advanced *tricks* in pole dancing, where a person's body is upside-down while they are gripping the pole. The necessary grips for these moves often require contact between skin and the pole, which wearables such as watches can interfere with. Sketches drawn by Carolina Arias Arguedas.

impacts how polers in turn participate in the practice. Here, we highlight how polers find that "feel" is a crucial element to their activity in order to trace the potential for polers as co-conspirators for pluralistic engagements with the design of self-tracking devices.

4.2.1 How polers participate in somatic reflection. "Feeling" is a primary aspect of pole. To feel, is to be able to intuit the minutiae of ones essence whether physically, emotionally, psychically, etc, at at any given moment. How one feels influences movement and is influenced by movement. Many described this relationship as a "mind/body" connection, and emphasized how central this practice is for poling.

Being aware of one's soma shows up in a myriad of areas. For example, in thinking about performing a standard inversion, P9 described how she faced challenges because she could not translate instructor descriptions of the movement for her ow body. It was only through personal (physical) reflections and trial and error, that she was able to discover how inverting can work *for her own body*.

> *"...I was struggling for a [regular] invert [for years]... And it wasn't until I started playing around with the different ways that I would grip the pole. The different ways in which that I would engage my back, and the different ways in which I can actually like tuck my pelvis underneath. Did I actually get it to click for* ***me*** *..."* [P9]

But this practice of feeling is not necessarily isolated to the moment one comes into contact with a pole, on the contrary, *intuiting* is a constant part of polers lives. To be a pole practitioner is to constantly partake in somatic awareness.

> *"...[pole] made me a lot more like forgiving of myself when I recognize the fluctuations in my own body, you know whether I'm just like from getting older each year, or from you know, traveling during the holiday, or, you know, having a really like painful period cycle, like whatever. It has definitely kind of made me recognize my own kind of like body patterns and responses. And yeah, just kind of be more accepting of my own expectations of my body."* [P8]

This attunement to how one feels is then also extended to how polers approach their physical practice and understand their capabilities once in the studio.

> *"...there are some times where I'm feeling a certain way, and you can see the way that I dance is different,* ***because it's an expression of me, and how I'm feeling****. So sometimes I'm tired. My body hurts or shit is going wrong, and I'm like, just very slow, and like I may do body rolls this whole time. But yeah. And then there's some times where I'm feeling good. I'm feeling strong. I'm feeling like a badass, and then I get up on the pole, do some like badass trick that takes a lot of strength..."* [P3]

Polers also frame feeling as multiplicitous. By noting that *feel*, and that what one can do as a result of said "feeling", in the practice of their activity, is constantly evolving, polers emphasize that ones' capabilities are not static. Even more so, the reasons behind these changing (felt) experiences are not always easily explained. Take P4 for example, who describes her that her relative flexibility varies, and the reasons for that variation at times is not immediately clear.

> *"I know I could be flexible. But* ***for whatever reason****, I'm not flexible today. So stiff in my body."* [P4]

Additionally it is important to emphasize that the way *change* manifests is not widely translatable [31]. Even more so, the assessment of change whether improvement, regression, or any other such combinations, lives in different areas of the body uniquely, and this reality is also constantly negotiated by polers.

One place this shows up in pole is through the asymmetry of the body. P9 describes the phenomenon of ability variations between the right and left sides of the body, as well as on the vertical axis.

> *I mean progress in pole is hard, because you know, bodies are asymmetrical, we all have dominant sides, like most of us are not ambidextrous for example. So I track my progress where I can like see on my non-dominant side how far I'm coming, you know I always attempt things on both sides..."* [P9]

4.3 RQ2: What are participant perspectives on the (physical) use of tracking devices for poling?

As we outlined in the previous section, there is a specificity to pole dance which requires precision in bodily placements but also a concerted inner relationship with the kinetic sensations of feeling. In this section, we highlight how polers understand the experience (or non-experience) of having a digital assemblage while doing an activity which has unique constraints and affordances.

4.3.1 Concerns around aesthetics & jewelry. Although many participants had used trackers, either for pole, or generally, there was also a contingent who either no longer used them or had never used them. Some polers remarked that they did not use trackers on the basis of artistry and aesthetics. For these polers, they felt that their artistic desire for fluidity, and aesthetically pleasing performances, could be hindered by the fact that commercial wearables are (commonly) watches worn on the wrist.

"I personally find devices. You know, like rings or watches, I find them quite clunky again, coming from like a dance background... it sort of cuts off the line. And I find aesthetic wise, it's not something that I like to see..." [P14]

Others may use the devices in their day-to-day lives, but not in their pole practice. Some of these dancers explained that they did not use the devices because the fit of the watch was prohibitive for fully engaging in contact with the pole 2.

> *"I know I had my Fitbit when I started pole, but I just took it off. Because part of me was like, I need my wrist free."* [P1]

Additionally, some users do not use the watches, or use them in adaptable ways, because of studio jewelry policies. These policies are wide ranging in severity, with some studios completely banning jewelry of any kind, including even earrings which dangle, to others only banning jewelry whose material could scratch the pole. Some instructors noted that not only do they not wear jewelry themselves in the studio, but that they make it a point to remind their students that they cannot wear them during classes.

> *"No, because I usually tell them to take off their rings and then their watch and everything. You know, that's kind of like one of the policies that [the] studio has. Because the rings will scratch, even including the watch, all will scratch the pole."* [P4]

However, jewelry policies for pole are *not necessarily* prohibitive for wearing fitness trackers. Many polers persisted despite these potential barriers.

4.3.2 The physical experience of wearing a tracker. Some polers who did use the device on the apparatus emphasized that any difficulties with wearing the watch on the pole either were uncommon, and/or had workarounds, such as through the use of watch bands with specific materials like those made of rubber. For instance, a few noted that they sparingly had incidents where using the watch caused potential danger.

> *"I got stuck, and I like kind of tore a piece of it [my watchband], trying to get my hand out. Otherwise I was gonna fall. So I was like, we gotta get this hand out between my legs. So I did get stuck on my 1st Garmin watch and I kind of tore the band a little bit trying to escape. Other than that one incident, I have not had any difficulties with wearing a watch while poling."* [P3]

Yet, although the watch is compatible with their (physical) movements more broadly, in the areas where it becomes more difficult, they remove it and carry on with their session. For example, for movements with grips that rely more intricately on the wrist, forearms, or are generally more difficult to perform.

> *"...I just put it on, but I do sometimes have to take it off for certain moves when I teach, because it gets in the way... when I teach, advanced pole that's when it really starts getting in the way of things, and I just take it off mid-class."* [P6]

Throughout the interview, P19, a full-time movement artist, expressed many positive experiences with her Whoop device, a wrist-worn commercial fitness tracker, which she uses both for pole, generally for a comprehensive outlook on her health, and for helping her manage her fitness goals. However, she recognizes that there are unique challenges to using the watch on pole. Here, she describes what we will describe in the discussion as a "balance" paradox, or the necessity for dancers to be competent on both sides of the body, meaning for doing tricks, or flexibility skills like splits.

> *"It's not ideal, because in reality we shouldn't be wearing like things on our wrist... we want to do everything on both sides as equal as possible. So what happens with Whoop... during a class or during my training I have to keep switching my Whoop from being on this wrist to this one depending on what I am doing. So it's like a lot of like putting it back and on, back and on. I've tried, they have like a sleeve, but it's worse because we use a lot of like the bicep and the armpit to do a lot of stuff. So it didn't work at all. So I just keep putting my Whoop on one wrist and the other and the other."* [P19]

On the other hand, having a watch was not only potentially useful for tracking personal data, but also for the more practical experiences of pole dancing, like changing music while teaching classes. Many polers expressed that the functions which connect fitness trackers to Bluetooth speakers were useful for limiting the physical back-and-forth they must do to change music from their phones.

For some, this was the primary reason they purchased the device.

> *"So mostly, I started using it because as an instructor I can use it as a remote control for the music..."* [P16]

This experience of having to trudge back and forth between phone and pole is especially cumbersome when we consider the fit of certain apparel items central to the practice, such as stilettos.

> *"...I like the idea of being able to control music at the speaker without having to get up and walk over to my phone when I'm in 8 inch stilettos, you know, I think it could potentially help with the flow of my training."* [P20]

4.4 RQ3: What are poler reactions to data that can be obtained from wearables in relation to pole dancing?

Tracking data for pole requires unique configurations which often times result in incomplete data due to the incompatibilities with the movement type. In this section, we highlight the ways that data is understood by polers and (if any) the workarounds they have developed to find value in self-tracking for pole.

4.4.1 Activity mode used in lieu of pole being available. As of the writing of this paper, pole dance is not available as an activity on any commercial fitness tracking device. Yet despite this, polers often found methods for gathering and recording their activity. Many have trouble deciding which "mode" to track their activity in. As discussed, pole dancing does not fit neatly into any activity types. The way that polers engage in their work is pluralistic, and can be a slew of different "modes" at once.

> *I feel like a lot of things in pole are more strength-based. But it doesn't translate, right? So like your strength training, you have to put in specific exercises for it to calculate your muscle load. But they don't have pole dance exercises on there. So you can't calculate that. So you just have to log it as dance, where I feel like it's so much more than that. So, yeah, I'm dancing. But I'm also like climbing up a metal pole and flipping my body upside down. So that doesn't quite translate...* [P17]

Participants often used the "dance" mode on their respective devices, despite feeling that it was not representative of what they do. Yet, because many polers purchased the devices because they had a desire to track their overall health, they felt that they still needed to log *something*. However, this diminished the value that they got from the data because they did not have a tangible knowledge of how to use what they tracked to understand their progress towards their health goals.

> *"I never thought it was accurate, because the only mode that I can really use it in is dance... so what I found was that... it didn't really change much, you know, or it didn't really help me in a way where I was like, oh, yeah, this means I'm improving. Or, yeah, this means that whatever like kind of insights you get from fitness tracking." [P16]*

Others were more particular about the modes that they use, going more towards the Circus Arts (on Whoop) or generic 'Activity' modes. Generally, the same issues arise. Here, the Circus Arts mode picks up on the strength parts of the practice, but does not translate the dancing portion of pole.

> *"...they have a mode that's called circus arts...it translates very good in the fact... like different muscle parts and the body engagement...(but) when they do circus art, they don't do as heavy dancing as we do, so it's still not accurate as I would love it, but like from everything that's out there right now, it's the best option, in my opinion."* [P19]

4.4.2 Incompatible data. There are other parts of pole which do not translate to what fitness trackers purport to track. These we feel are akin to intangibles. Take for example how central flexibility is to the practice. In flexibility classes, students perform both static and dynamic stretches, which for the former, your heart is closer to its resting rate and the latter elevates it gradually. Some polers did not wear the fitness trackers for pole precisely because they felt it could conceptualize these nuances.

> *"No, it's like, how do you track flexibility on a watch? I don't know how that works."* [P5]

For those who did track for flexibility classes, polers perceived that the devices had trouble tracking the intensity of the activity. This made the polers question why this was the case, for them,

flexibility, and by extension, pole, should elicit the same acknowledgment as physically intense practices just as one would other activities.

> *"I also do a lot of like flexibility training**... Whoop was not considering flexibility as an activity. Oh, you're just stretching. No, no, I am sweating. My muscles are shaking. So that was annoying for me because I was like, this is a workout. I'm fully working out.** Maybe my heart rate doesn't go as high but I am fully working out..."* [P10]

And for others, the trackers would simply not track the activity.

> *"...I used to track when I did flexibility...even when I was teaching flexibility, it would think I'm not doing it..."* [P6]

But this phenomena was not limited to flexibility. When doing tricks, the tracker still does not accurately acknowledge efforts which the polers consider high. Thus, dancers may only set the tracker for classes which may be more cardio intensive because the trackers capture those movements better.

> *"...I'm not expecting big results from a tricks class. I would think to set it if I was doing a twerk class, because I know I'm about to be doing cardio and jumping jacks. But in my experience weightlifting and tricks I don't think it's gonna be an accurate result of how hard I'm working. Cause I'm working my ass off, and my arms are shaking. But it's gonna be like, okay, you did a little something like, no, I didn't. I was hanging upside down by my pinky toe and it's not going to acknowledge that as a pat on the back."* [P3]

In thinking about the translation of step count for an activity that does not take steps, polers also described how they conceptualized their effort in comparison to the effort required for a step. By reflecting on the data that way, they find a gross misrepresentation of their efforts. For some, this made them discontinue tracker use for pole.

> *"...it's not designed for this type of movement. It's designed to count steps, so we'll be doing very hard physical activity, and I have worn it, and it's measured like 20 steps for the hour, and I know I did more than 20 steps of moving. It's not accurate at all. And when I realized that fairly early on, I was like, Let's just take it off because it kind of was like, who cares if it's not actually measuring anything I do?"* [P11]

The tracker also has difficulty engaging with multiple modalities within one activity session. Meaning that transitioning from doing more strength based movements, to more dance based, is not very well understood.

> *"It has a hard time tracking the part of moving from like being on the pole doing tricks to like you're on the floor dancing, but that does not mean that you're doing nothing."* [P19]

In reflecting on the use of personal trackers for pole, some participants made it a point to discern that the somatic connection to the pole work was still necessary. They noted that because of the device's limitations on what returns about the body it can provide, some do not fully engage with the data.

> *"Because, unfortunately, this device cannot like tell you everything. Oh, your muscles are this fatigued? You know there's some limitations to it, so I feel they're useful but I don't go like oh, I trust them 100 percent. This is all I'm believing."* [P10]

Others were more explicit in stating that the somatic understanding of ones body cannot be, and should not be, replaced by merely looking at fitness tracker data.

> *"... it's very difficult if you're a newbie to to be able to discern. How is your body feeling? It took me a long long ass time to, understand, like, you know, signs of over training signs of exhaustion ... our bodies are not like machines... device trackers are helpful, for sure. I definitely felt sometimes I felt like crap, and the device was confirming that I was feeling like crap like everything goes down... I'm not saying, everything is just bullshit... But I'm also gonna listen [to] what my body has to say."* [P9]

Or in ideating possibilities for tracking returns outside of the conventional metrics, *some*, were specific in stating that they were not interested in tracking core pole movements like strength.

> *"...you can tell if you were stronger than the day before, just upon like, okay, I got higher or I held something longer. Others might have a different opinion. I don't really care if a device tells me I'm stronger.* ***I'm gonna feel that****."* [P11]

4.4.3 Hacking. In order to service their desire to know their fitness data, participants partake in processes of "hacking", wherein they adapt their needs to what is available, and fill in the gaps on their own.

For some participants, particularly those who expressed an interest in data tracking for health concerns, they merely wanted a birds eye view of their data to generally gauge their weekly physical activity levels.

For them, some data was better than no data, even if that data was partial or incomplete.

> *"I mean, well, I feel like it's still gonna track some kind of movement, right? So it's worth it for that."* [P17]

By logging the activity, even if the data is inaccurate, they were able to have a record of their previous exploits, which when looking back on their activities over time, was useful for generating health insights.

> *"Honestly, ultimately for me, I just wanted to see a really high-level overview, to see how my week was trending when I would look at my movement activity, or fitness, activity. I would just look as like just a quick week, and I would look at the month. I wonder how much activity I did last month. I would go back, and I would look to see if there was any patterns or trends there."* [P13]

This method is also useful for helping to decide the activity load they would take on for the days following.

> *"...I just do it to keep track of things. And just to say, like, 'Oh, my God!' Today I really did so much, you know, and maybe tomorrow, I'll take it a little bit lighter.* ***It's more for me to have an idea what's happening.****"* [P10]

Others were more specific in their data analysis. For example, P7 described using the heart rate data collected from her practice sessions to help her discern how much she was improving on her routine. P7 participates in competitions, and within this space, polers are expected to perform a standalone choreographed piece for a length of time determined by the rules of that particular competition. Regardless, dancers must have the stamina to be able to complete a routine from start to finish. Using the device, she was able to unpack her stamina (in some part) by connecting it to the declining trend of her heart rate data.

> *"... related to using fitness trackers, one thing that I was looking at like as I've been working on my competition piece is like when I'm doing these full run throughs. What like my max heart rate looks like each time. And it's kind of cool like it actually has like gone down like slightly every time that I've done a full run through"* [P7]

P19 had the same technique as P7. Using her Whoop device, she tracks pole using circus arts, and then analyzes the heart rate "Zones" data point disseminated by Whoop, to help gauge how she had improved her stamina when performing her competition choreography.

> *"... it's not the same doing one trick and coming off the pole, and like maybe resting and like drinking water, but having to do 4, 5 min of that without stop, and making it look pretty and like it's nothing...* ***It's really key for me the zones that I'm working. If I'm doing my routine and I'm in zone 5 my entire routine but I've been doing it for a month, and now I can do it in a zone 3 or zone 2. That tells me, okay, my body is more capable now...****" [P19]*

5 Discussion

From this work, we found an impractical but negotiable relationship between polers and fitness trackers. Because the practice consists of a set of different movement modalities, the manner by which polers are able to engage in fitness tracker use offers unique tensions. These use cases are affective both on a physical level, i.e, the experience of wearing a watch or ring for a practice which requires skin on skin contact, and on the data level, i.e., what value can be derived from data analysis. We found that although pole dance is not an option on prominent commercial tracking devices, some polers appropriated the data that is available to serve their individual needs. Although the fit of the watches is not ideal, some polers are able to persist in wearing them for pole dance. Even still polers contend with the limitations of such tools in capturing the dynamicity of their movements, temporal change and its impacts on movement, and generally call attention to how the act of "feeling" is not necessarily something which can or should be relegated to a device.

It is important to note that we are not working within the confines of a "representational" politic(s) or "inclusion" feminism, which suggests introducing the excluded into an already existing system. This politic, although on the surface appearing to be transformational, does not transform, because it does not require a reconstruction of the systems they attempt to enter. If we were to work from a representational standpoint, the obvious conclusion from the findings would be to suggest that designers include pole as a mode on fitness trackers. This is the easiest answer, and it is not necessarily inaccurate from the data that was collected. For many of the polers we talked to who are interested in data tracking, this is an actual desire. However, this does not get to the heart of this work. Such a politics does not allow us to re-frame, negotiate, and ultimately re-articulate, what we hold constant as a discipline and what we do not hold at all. Instead, we recall postmodern feminist theory and its exploration of pluralism [26]. Pluralism rounds, it asks us to contextualize people and their interactions with the world as ever-evolving in their complexity.

From talking with polers we found several distinctions of the practice, which displace it from the universalist assumptions of self-tracking devices, and in unpacking the participants persistence (or resistance) in using the devices, we offer a discussion about self-tracking and pluralism in movement and feel.

5.1 Expanding sensing type

Gripping, squeezing, climbing, and more are essential physical movement types for pole existing within a larger framework which generally relies on strength, flexibility, and artistry. In their descriptions polers emphasized how these fundamental techniques require complete use of the body and most importantly dry, uncovered skin. Skin then, is also a form of "technique" in that sincere considerations regarding its interactions with the pole and the clothing or jewelry one may wear, become essential for how polers can use trackers.

Therefore, reflecting upon the physical requirements of the practice, we offer inquiries for designers around the kinds of sensors we use in our devices and where we place our wearables.

5.1.1 Sensor type. We argue that properly indicating polers range of motion cannot be easily reduced to *any* individual movement types. If the wrist based tracker was able to sense grip strength, for example, this would not be exhaustive enough for the breadth of movement within the practice.

Take for example a three minute routine choreographed for a class. A poler moves their waist, swirls their head with a flourish, dips low onto the ground, they're dancing, but just as quickly, they could be crawling towards the pole, and with a dramatic flair grabbing it with their right hand and slowly standing up before hooking their right knee pit on the apparatus, then spinning for a few cycles. They may then come to a stop momentarily, so they shake their hips to the side in a sensual manner, before wrapping both of their legs around the pole and using both arms and legs climbing vertically upward before sliding back down lower to the ground. In the matter of three minutes, they have activated movement and all of its potentialities. A dancer can go from being on the pole, and that being their primary apparatus for expression, or merely being around a pole without touching it.

This is *flow*.

Polers are also operating on both vertical and horizontal dimensions and there are long stretches of time where polers do not come in contact with the ground at all. Consistently, they contend with gravity both on the vertical plane and the horizontal. How do devices contend with gravitational force in calculating exertion? Such physical concerns are similar to other aerial sports like gymnastics, or field sports in track, or even manual labor activities such as window washing.

As designers, this presents a conundrum. Such dynamicity produces motion and strain that cannot merely be evaluated through heart rate sensing and surely not step count tracking. There are other forces at play, both literally, in thinking about motion and gravity, and metaphorically, from a more critical theoretical standpoint.

This breadth in variation is what we call *sensing zones of plurality*. Because digital tracking technologies assume a fixed tracking modality, owed to the limitations of sensor types within the watches, the wearable only senses specific sensations of the body at any given time. This becomes problematic for a dynamic activity like pole, because there are different sensations occurring at the same time which impact the overall expenditure of the poler. The physical requirements do not all necessarily have ties to biometrics. Although we describe wearables as residing in completeness, in actuality they reside in the nominal. The *way* that this contention shows up in pole is unique to pole's affordances, but this problem of incomplete and incapable sensing is widely reported in a plethora of physical activity types outside of running [17, 29].

Some of this tension can be traced to the type of sensors used in the device and the placement of those sensors on the body. Zeagler contends that sensor placement is one of the most important considerations for gathering valuable data, and identifies that where one places a tracker, best placed on the arms, hands, waists, lower backs, and head, should be in relationship to the most essential nature of that activity[68]. There have been attempts to reorganize sensors to other areas of the body like gyroscopes and accelerometer sensors placed onto the ankle for jazz sensing [58], but in thinking about pole, and other such activities which are dynamic, singular sensing areas do not prove optimal. In this practice,there are opportunities to be very particular, as there are even "stretch" sensors, which would prove invaluable for flexibility tracking [68], but there is an overall need for a sensing ecosystem which accommodates for a more comprehensive and accurate outlook.

5.1.2 Form. Although some polers used watches while poling, all held the opinion that the wrist was not an ideal place to have their trackers. For some, the watch negatively impacted the aesthetic value of their movements, for others, it impeded full access to their wrists, particularly for grips. For those who were able to continue with use, some still expressed that there were advanced tricks that they had to take their watches off to perform. Polers were not able to use ring-based fitness wearables for safety reasons.

Similarly, watch placement came into consideration when discussing the *balance paradox*, where polers strive for consistency between their left and right sides, and below and above the waist.

This balance paradox, although not described using that term, is a known phenomena in pole dance, and there is a known imbalance between the grip strength of polers right wrist (most common dominant side) and the left wrists [45, 46]. In a class session, it is not uncommon for instructors, to the response of groans, to direct students to try a trick or a dance movement on the opposite sides of their body. Thus a movement in pole is not always performed with equilibrium on both sides, making it as important to evaluate your relative capabilities on both sides of the body. This means that for some polers who were interested in gathering more intricate data about their relative abilities on either side, they had to switch which wrist they wore their watch on.

This then opens up the opportunity to question the ubiquity of the fitness tracker "watch" iconography. Some could say that the watch is most useful because it is where we can best collect biometrics like heart rate. But there are at least nine pulse points on the body, including parts of the feet, popliteal (inner knees), femoral, and other such areas which are pivotal zones for pole movements [43]. Having a watch, which a person is expected to wear on one side of their body, where it then gathers "complete" data, for activities which have the balance paradox, cannot gather the nuances of exertion as they change between dominant and non dominant sides.

The decision to have a watch is not easily reconciled to the fact that the wrist is a pulse point. Rather, we are inclined to believe that it reflects a body politics that Foucalt describes as power, which is something not intrinsically synonymous from a top down level, whereby systems impose ideas onto individuals and thus individuals develop cultures and identities around ideas. Rather, power is "omnipotent" and constructed and replicated by way of social acceptance [42]. It follows then, that since design is is founded in masculinist ideals, and wrist watches are popular and marketed symbols of masculine desirability and social status, the most popular form that a self-tracker takes could be a watch [15, 55].

Yet, the wrist watch is inadequate for pole and other types of activities which rely on extensive use of the skin, because it not only gets in the way, but its form has no utility outside of designer convenience. Knee pads are common in pole dance, as are bikini bottoms, bras, and heels, and are located in places both identified as most optimal for sensing movement, and pulses, for those who are invested in heart rate data as well. We suggest evaluating the role form plays in design, particularly through the use of textile wearables [51, 69]. In ideating within that realm, it is important to consider other factors such as cost [36], and re-usability considering waste from device abandonment [66]. Creating wearable fashion both disrupts the aesthetic displeasure some polers may have with wearing watches but also frees the body to move as it usually does. Embodying a more expansive imaginary, we turn to what Turmo Vidal et al. discuss about the "wearer" of the technology as an important dimension to consider in the design of wearables [61], thus we leave room to explore removing the assumption that the wearable must be used on a body or bodies, and not necessarily used *by* a body, i.e. a pole as a tracker.

5.2 Feel

Polers also expressed how important the feel of the body was to how they dance. Feeling is central both because pole is an artistic discipline which predicates using movement to show and elicit

emotion, and one which requires a staunch awareness of the way ones body moves technically. Towards the latter, changing emotions, stemming from social or psychological experiences at those times, impacts the way their bodies move. But physiological or physical concerns of the body, such as the phase of one's menstrual cycle, or stage of bruising on skin, also defines the physical range within which they can play. And although not extensively enumerated in this paper due to the breadth of generative findings, feeling was not only relegated to those difficult or challenging aspects of the polers' experiences. They were able to feel immense joy, peace, sensual and cosmetic, pleasure from pole, and these too came to light in their embodied experiences. To service this aspect of their activity, polers emphasize that building their muscles of perception is key.

For polers who were interested in tracking, there was a general consensus that this felt experience was still valuable and should not be delegated to a tracking device. However, polers emphasized that they still wanted watches to recognize these temporal experiences of feel when calculating or returning data. This came up most often when discussing the menstrual cycle effects on both physical and emotional status, and aging.

When poling, participants described feeling the aging in their bodies. Some had been dancing for over a decade, all of them under the age of fifty, but still their bodies feel different than when they started. Some had given birth, which could explain some of the difference, others have injuries, newly diagnosed disabilities, etc, but regardless, all have gotten older. Within HCI, aging seems mostly to apply to persons above a certain age threshold [63]. From our vantage point, fitness trackers do not directly acknowledge aging as a more universal experience (considering all the design emphasis on universality). This is particularly demonstrated in the static daily "fitness" goals which users are encouraged to set. The expectation that one shall walk 10,000 steps a day, every day, for example, removes feeling. And outside of the wrist-worn Whoop device, which P19 described as giving her changing daily goals based off of her calculated "strain", none of the other polers had a relationship to the data on their watch that was inclusive of these realities. For our exploration into device abandonment, we should also consider how, if we aspire to design devices which are used for extended stretches of time, data recommendations, representations, and calculations will need to adapt to the plurality within the body, both due to, and irrespective of, pole dance, especially as we continue to age.

We surmise that personal informatics should not necessarily rely on thinking on the "body as anatomy and physiology", which as we see flattens the body as a site for de-contextualized data to come out of, but instead think upon the mind/body in personal informatics as "knowledge" and "expression" [39]. Although these two concepts offer different standpoints, in aspiring for a pluralistic design space, working with the idea that "the body can be used as medium to think with" but also that which acknowledges that "the body is visible and material, an expression of vitality, but also a mode of communication with the world,"[39], and even that to be connected to one's body is to constantly be in cycles of "failing beautifully" [59], opens us up to the possibility of translating the inextricable relationship between the body and complex *feeling*. Steering clear of a techno-solutionist impulse to relegate all feel to a tracking device, we ask instead, **What does it look like, if even possible, to implicate ever evolving felt experiences in conversation with physical activity tracking (for pole), that is not necessarily based in diarying or journaling?**

One way we can move towards answer this question is through a concerted centering of women in fitness tracker design. As Bardzell argues, centralizing women in our design space is an acknowledgment that the unique insights women, as a result of their societal entanglements with power and their individual experiences generally, "hold and produce different types of knowledge from their male counterparts, and that these different types of knowledge should be recognized and utilized as a resource rather than marginalized [11]". Polers specifically have much to offer our discipline as we continue to think through embodiment and wearable technology. This is precisely because of polers insistence on understanding the body, the unique methods they go about understanding it, spanning all manners of movement from the sensual, to the aerial, to the breath and more, and the complex circumstances that arise from using a pole which elicit particular framings of the body and device use (outlined earlier).

But also, pole as a somatic activity which engages the erotic, can be a useful mechanism for re-envisioning the tracker model as a site which *reveals* from "outside ourselves", to a model that "lives[s] from within outward" [40]. **What would a device framed to sense and return data from a computational and theoretical standpoint that *lives from within outward* be like?** As we hope to move toward a pluralistic personal informatics, pole dancers as somatic practitioners can be of immense value as *co-conspirators* in our pursuit of these realities.

5.3 Data Goals

While polers often did not find relevance in the data produced by wearables, many persisted, finding ways to manipulate what was available for their own purposes. For those who were only interested in tracking to keep a record (e.g., documentary tracking [23, 52]), the inability to track pole was not a major barrier. For those who were more intent on data analysis, they had creative value-seeking methods involving the appropriation of heart rate data which they leveraged to recognize their improvements in stamina. Some hoped to understand changes in flexibility (although not available), a few to keep a health log, others simply to change music, and unexpectedly to track more normative fitness information like caloric expenditure.

Reflecting then on Turmo Vidal et al.'s design framework for wearables, any work done within this space to address these desires would first have to begin by defining the goal(s) for wearable use in pole, are they, (not exclusive) to be used for improvement in the practice, to gather health metrics, or maybe to generate some other possibilities that are not yet apparent? Either way, by establishing our goals, we can begin to articulate more specific characteristics related to the input, output, and temporality of device use, i.e. "*when to include them, and when to use them*" [61]. Because of the role that presence and fluidity play in the practice, however, fluidity in input, or what is tracked, as already mentioned, will be one of the most crucial dimensions. Therefore, we could ponder the generalist design of wearables, which assumes a fixed range of goals, or that the objective of goals is physical "improvement" and instead,

design for a multitude of objectives, allowing users to select their motivations, thus creating an opportunity to refine their output.

We stop short of offering specific *specific* design suggestions related to "output" [61] for pole because in keeping with the tradition of soma design, such suggestions must come from embodied methods, wherein ideation happens with the self in practice and not intellectual isolation. In other works, soma design has been used to conceptualize data into embroidery [32], teddy bears [8], warmth [24] and even the body itself [50]. In our context, we can use soma design methods to approach inquiries on how polers envision the representation of data.

Additionally, future work in this space will require that polers take the role of experts [31], a move that has been heralded for the possibility of the breadth of discoveries, like what was generated by the role of "sexperts" brought on to assist in the design of sex toys [10]. This not only allows for a set of generative investigations for the personal informatics research area, but pole practitioners can also inversely gain a deeper understanding of tracking for themselves. And in keeping in line with pluralism, we emphasize that this future research work builds upon what Dourish & Mainwaring [20], have articulated that the goal is not to design for a *globalizing* context, but a local, situated one, that which rejects the notion that there are observable attitudes, feelings, and experiences that can be reproducible across "scientifically validated user types" [11]. Therefore, the future research site(s), whether the strip club, pole studio, pole competition, or cypher is not the means but the end.

Lastly, in considering the potential for pole dance as a site for soma design inquiries, we emphasize that many of the contentions and considerations we surface in this work can, in some part, be applied to other non-normative (to personal informatics) activities. We invite researchers to explore what we unpack here as it applies to the pursuit of surfacing generative self-tracking designs for their activities of interests. We call particular attention to the conversations we have on the importance of skin contact for performing pole and how that interferes with tracker placement, a design inquiry which could be explored in studies on gymnastics, manual labor activities such as dish washing, rock climbing, and others. There are also potential applications on the impact of vertical movement, and the relationship of gravity and how they could be corresponded with in designing for activities with gravitational considerations such as pole vault. Generally however, we hope designers can find value in reflecting on design as it relates to tracker placement, sensor type, and how persons at their activity site engage in the practice of embodied feel.

6 Conclusion

In talking with polers about their practice and their use of self-tracking devices, we were able to bring their unique perspectives into the personal informatics conversations. Some polers succeeded in making use of data from wearables, but many had challenges with the physical experience of wearing a device and what they perceived to be a general lack of useful data. We find that to begin to engage in a design process which acknowledges the inherent dynamicity of pole dance and other such complex movement practices, we should look to expand upon how we commonly think about sensing, both its material form, and the data it outputs. To begin to ideate this space, we suggest a re-imagining of tracking possibilities centered in principles of soma design.

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